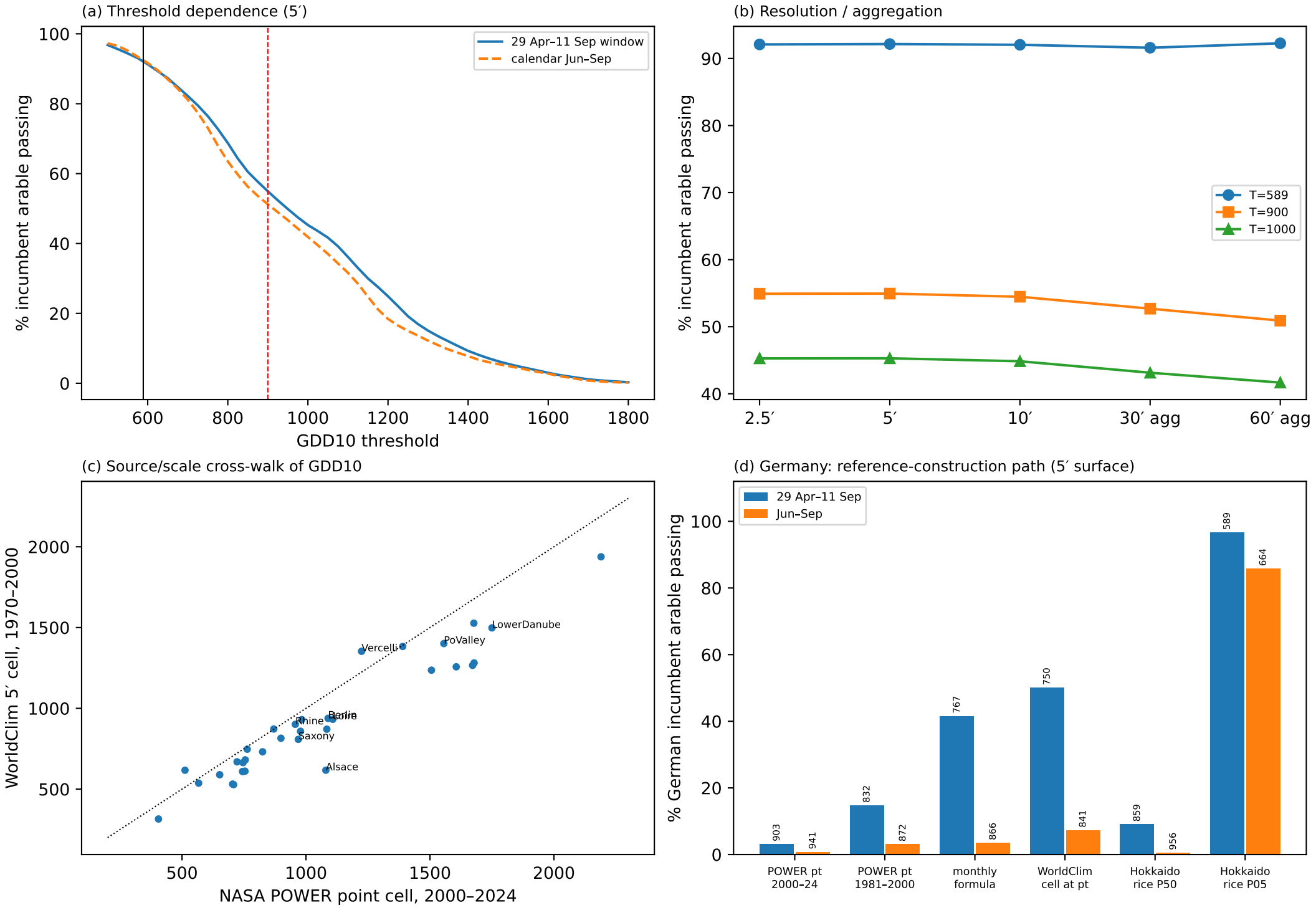
(a) Threshold dependence (5′)
% incumbent arable passing
GDD10 threshold
29 Apr–11 Sep window
calendar Jun–Sep
(b) Resolution / aggregation
% incumbent arable passing
T=589
T=900
T=1000
2.5′
5′
10′
30′ agg
60′ agg
(c) Source/scale cross-walk of GDD10
WorldClim 5′ cell, 1970–2000
NASA POWER point cell, 2000–2024
Vercelli
PoValley
LowerDanube
Rhine
Saxony
Alsace
(d) Germany: reference-construction path (5′ surface)
% German incumbent arable passing
29 Apr–11 Sep
Jun–Sep
903
941
832
872
767
866
750
841
859
956
589
664
POWER pt 2000–24
POWER pt 1981–2000
monthly formula
WorldClim cell at pt
Hokkaido rice P50
Hokkaido rice P05